

ASSIGNMENT 4

QUESTION 1

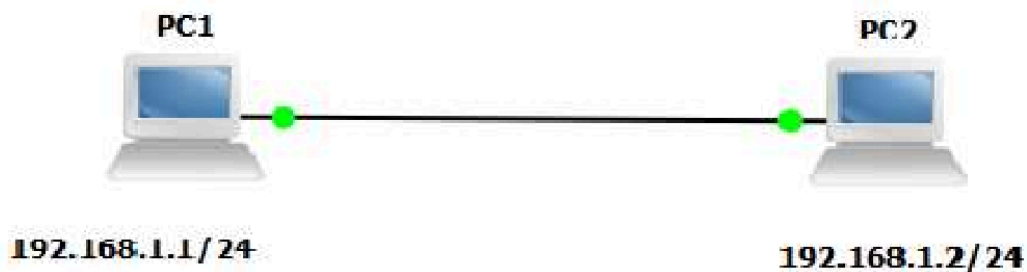
Discuss the GNS simulation environment.

SOLUTION

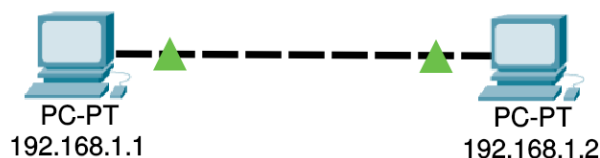
The GNS (G) simulation environment is a powerful tool designed to facilitate the simulation of various network protocols and systems. It provides a flexible framework that allows users to create, configure, and run simulations of network scenarios. The environment supports a wide range of protocols, including TCP/IP, UDP, and others, enabling users to model complex network behaviors.

QUESTION 2

Connect one PC with another PC as per diagram below.



SOLUTION



Ping Results (from 192.168.1.1)

```
C:\>ping 192.168.1.2
```

```
Pinging 192.168.1.2 with 32 bytes of data:
```

```
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 192.168.1.2:
```

```
    Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
```

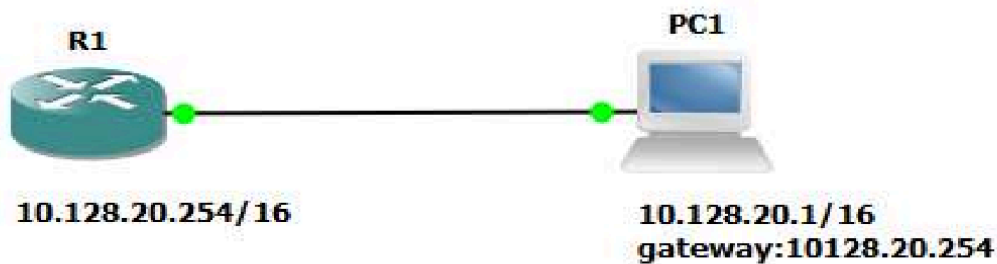
```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

txt

QUESTION 3

Connect a PC with Router Ethernet port as shown below.



SOLUTION



Simulation Results

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Del
	Successful	Router5	10.128.20.1	ICMP		5.000	N	0	(edit)	

Ping Results (from 10.128.20.1)

txt

```
C:\>ping 10.128.20.254
```

```
Pinging 10.128.20.254 with 32 bytes of data:
```

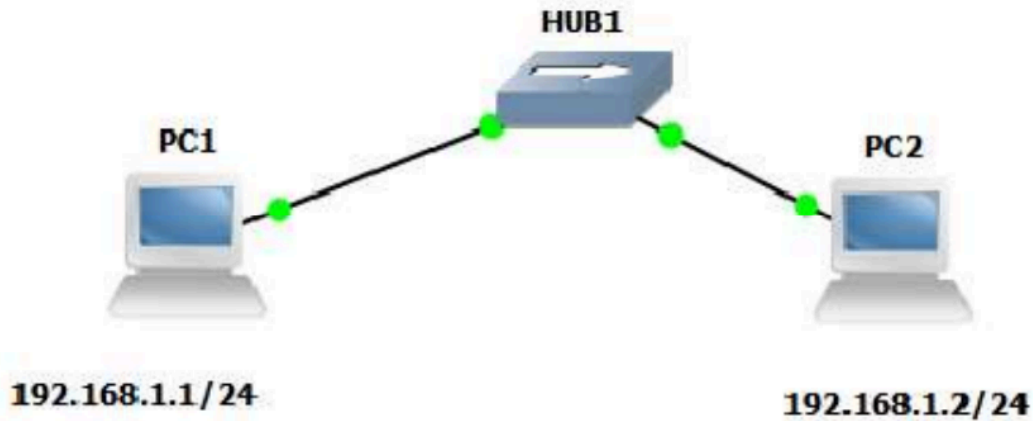
```
Reply from 10.128.20.254: bytes=32 time<1ms TTL=255
Reply from 10.128.20.254: bytes=32 time<1ms TTL=255
Reply from 10.128.20.254: bytes=32 time<1ms TTL=255
Reply from 10.128.20.254: bytes=32 time<1ms TTL=255
```

```
Ping statistics for 10.128.20.254:
```

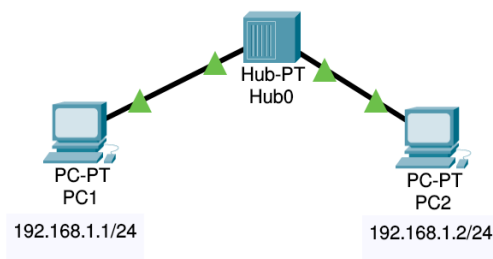
```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

QUESTION 4

Create a network of 2 PCs as shown below.



SOLUTION



Ping Results (from 192.168.1.1)

txt

```
C:>ping 192.168.1.2
```

```
Pinging 192.168.1.2 with 32 bytes of data:
```

```
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
```

```
Ping statistics for 192.168.1.2:
```

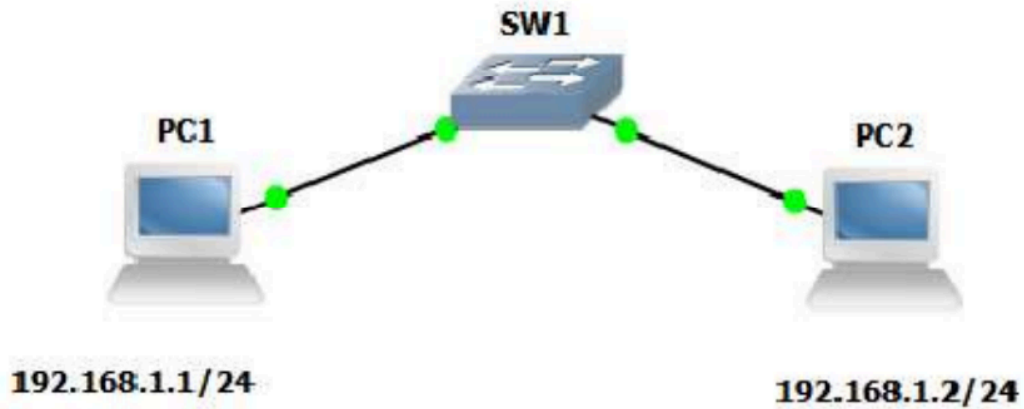
```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

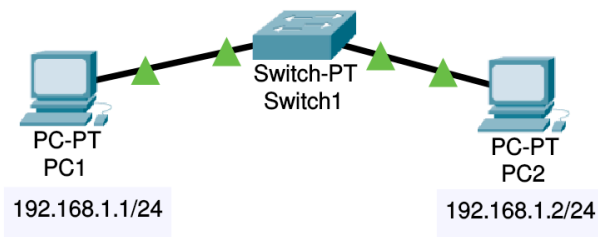
```
Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

QUESTION 5

Connect 2 PCs with a Switch below.



SOLUTION



Simulation Results

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	PC1	192.168.1.2	IC...		1.000	N	0	(e...	

Ping Results (from 192.168.1.1)

txt

Query successful

C:>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=1ms TTL=128

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Reply from 192.168.1.2: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.2:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 1ms, Average = 0ms